

Homework 6

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Problem 8

Problem 8.1. Write $\mathcal{R} = C^0(I, \mathbb{R})$ for I the unit interval. Defining addition and multiplication pointwise,

- show that $(\mathcal{R}, +, \cdot)$ is a commutative ring,
- determine all zero-divisors of $\mathcal{R}$,
- and compute $\mathcal{R}^\times$.

Proof of part 1. The main parts of this problem are the proofs of continuity. Let $f, g : I \rightarrow \mathbb{R}$ be continuous. We will use the fact that $+$: $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is continuous, and that $\mathcal{T}\text{op}$ has finite products (all of which I have proven in the past). The function $f + g$ arises as the composite $+$ $\circ$ (f, g) in the diagram

$$\begin{array}{ccccc}
 & & \mathbb{R} & & \\
 & \nearrow f & \uparrow & & \\
 I & \xrightarrow{(f,g)} & \mathbb{R} \times \mathbb{R} & \xrightarrow{+} & \mathbb{R} \\
 & \searrow g & \downarrow & & \\
 & & \mathbb{R} & &
 \end{array}$$

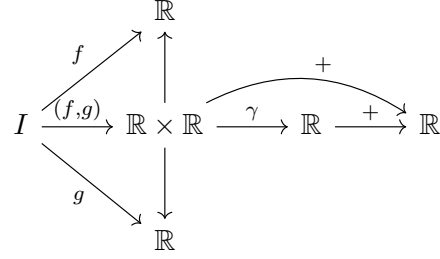
for $(f, g) : I \rightarrow \mathbb{R} \times \mathbb{R}$ the arrow arising by universality of product. Since $\mathcal{T}\text{op}$ has finite products, by universality of product (f, g) is continuous. Therefore, the composite is continuous, and $f + g \in \mathcal{R}$, meaning $\mathcal{R}$ is closed under addition. We will also use the fact that $\cdot$: $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is continuous. The function $f \cdot g : I \rightarrow \mathbb{R}$ similarly arises as the composite $\cdot$ $\circ$ (f, g) in the diagram

$$\begin{array}{ccccc}
 & & \mathbb{R} & & \\
 & \nearrow f & \uparrow & & \\
 I & \xrightarrow{(f,g)} & \mathbb{R} \times \mathbb{R} & \xrightarrow{\bullet} & \mathbb{R} \\
 & \searrow g & \downarrow & & \\
 & & \mathbb{R} & &
 \end{array}$$

We write $\bullet$ in the diagram to denote multiplication to improve readability. By the same logic, $f \cdot g$ is continuous, and $\mathcal{R}$ is closed under multiplication.

The function $\gamma : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R} \times \mathbb{R}$ mapping $(x, y) \mapsto (y, x)$ is a homeomorphism. In particular, it is continuous. The statement $x + y = y + x$ for $x, y \in \mathbb{R}$ can be rephrased as $+$ $\circ$ $\gamma = +$, since the left side is $(+ \circ \gamma)(x, y) = y + x$ and the right side is simply $x + y$.

We would now like to show that $f + g = g + f$. Indeed, since $+ = + \circ \gamma$, note that the diagram



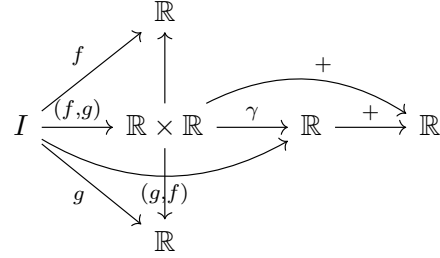
commutes. Therefore,

$$+ \circ \gamma \circ (f, g) = + \circ (f, g) = f + g.$$

Additionally, note that for any $x \in \mathbb{R}$,

$$(\gamma \circ \circ (f, g))(x) = \gamma(f(x), g(x)) = (g(x), f(x)) = (g, f)(x).$$

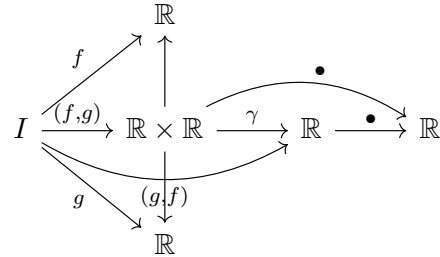
In particular, the diagram



also commutes. Therefore, we have the equality

$$f + g = + \circ (f, g) = + \circ (g, f) = g + f.$$

Since this is a composite of continuous functions, addition commutes. Similarly, note that $xy = yx$ in $\mathbb{R}$, so $\cdot \circ \gamma = \cdot$ as well. The same logic yields the commutative diagram



which witnesses the equality

$$f \cdot g = \cdot \circ (f, g) = \cdot \circ (g, f) = g \cdot f.$$

Therefore, multiplication commutes.

We will now show associativity of $+$ and $\cdot$. Let $f, g, h : I \rightarrow \mathbb{R}$ be continuous. Then since addition is associative in $\mathbb{R}$, we have for any $x \in \mathbb{R}$,

$$(f + (g + h))(x) = f(x) + (g + h)(x) = f(x) + (g(x) + h(x)) = (f(x) + g(x)) + h(x) = (f + g)(x) + h(x)$$

$$= ((f + g) + h)(x).$$

Similarly, since $\cdot$ is associative in $\mathbb{R}$, we have

$$(f \cdot (g \cdot h))(x) = f(x)(g \cdot h)(x) = f(x)(g(x)h(x)) = (f(x)g(x))h(x) = (f \cdot g)(x)h(x) = ((f \cdot g) \cdot h)(x).$$

Therefore, $+$, $\cdot$ are associative.

Now we show the existence of identities. We claim that $0 : I \rightarrow \mathbb{R}$ given by $x \mapsto 0$ is the additive identity, and $1 : I \rightarrow \mathbb{R}$ given by $x \mapsto 1$ is the multiplicative identity. Indeed, note that $0, 1 \in \mathbb{R}$ are the additive and multiplicative identities, respectively, meaning for any $x \in \mathbb{R}$,

$$(f + 0)(x) = f(x) + 0(x) = f(x) + 0 = f(x),$$

$$(f \cdot 1)(x) = f(x)1(x) = f(x)1 = f(x),$$

and the converse to both cases follows by commutativity. It suffices to show that $0, 1$ are continuous. Recall that if X has the discrete topology, Z has the trivial topology, and Y is any space, then all maps $X \rightarrow Y$ and all maps $Y \rightarrow Z$ are continuous; the preimage of any map $X \rightarrow Y$ is a set and is necessarily open in X , and the preimages of $\emptyset$ and Z under $Y \rightarrow Z$ are $\emptyset$ and Y , both of which are always open. Note that the singleton $*$ admits a unique topology, and this topology is both trivial and discrete. Write $0 : * \rightarrow \mathbb{R}$ for the map $* \mapsto 0 \in \mathbb{R}$ and $1 : * \rightarrow \mathbb{R}$ for the map $* \mapsto 1 \in \mathbb{R}$. Note that $0 : I \rightarrow \mathbb{R}$ and $1 : I \rightarrow \mathbb{R}$ arise as the composites

$$\begin{array}{ccccc} I & & & & \mathbb{R} \\ & \searrow 0 & & \nearrow 0 & \\ & * & \xrightarrow{0} & & \\ & \nwarrow 1 & & \nearrow 1 & \\ I & & & & \mathbb{R} \end{array}$$

where the map $I \rightarrow *$ is the unique map to the terminal object in $\mathcal{T}\text{op}$ (the same as in Set). Therefore, $0, 1$ arise as composites of continuous maps, and are therefore continuous.

Finally, we show inverses exist. Given $f : I \rightarrow \mathbb{R}$ continuous, define $-f : I \rightarrow \mathbb{R}$ as the map $(-f)(x) = -f(x)$. This is clearly an inverse since for any $x \in \mathbb{R}$,

$$(f - f)(x) = f(x) - f(x) = 0, \quad (-f + f)(x) = -f(x) + f(x) = 0.$$

It suffices to show continuity. Indeed, $-f$ can be written as the product $(x \mapsto -1) \cdot f$, and by continuity of f it suffices to show that $x \mapsto -1$ is continuous. This arises in the same way as 0 and 1 : writing $-1 : * \rightarrow \mathbb{R}$ as the map $* \mapsto -1$, we have

$$\begin{array}{ccccc} I & & & & \mathbb{R} \\ & \searrow -1 & & \nearrow -1 & \\ & * & \xrightarrow{-1} & & \\ & \nwarrow -1 & & \nearrow -1 & \\ I & & & & \mathbb{R} \end{array}$$

Therefore, $\mathcal{R}$ is a ring. ■

Proof of part 2. Let $f : I \rightarrow \mathbb{R}$ be nonzero, and vanish on at least one $x \in I$. Write $\text{Ker } f$ for the set of all $x \in I$ with $f(x) = 0$. It's simple enough to construct a function g with $fg = 0$; simply define $g(\text{Ker } f) = 1$ and $g(I \setminus \text{Ker } f) = 0$. The issue is making g continuous. Our strategy will be to construct a function g in a piecewise manner by defining $g(I \setminus \text{Ker } f) = 0$, and then having it go up and down while $x \in \text{Ker } f$ to have $g(x) \neq 0$ in a continuous manner. We will then see from this construction what extra conditions f needs in order to have a zero-divisor.

We claim that $\text{Ker } f$ is closed. Since $\text{Ker } f$ is the fiber of 0, and $\{0\} = [0, 0]$ is closed in $\mathbb{R}$, it suffices to show preimages of closed maps are closed under continuous maps. Given $g : X \rightarrow Y$ a continuous map, let $Z \subseteq Y$ be closed, meaning $Y \setminus Z$ is open. Therefore, $g^{-1}(Y \setminus Z)$ is open. It suffices to show $g^{-1}(Y \setminus Z) = g^{-1}(Y) \setminus g^{-1}(Z)$. If $x \in g^{-1}(Y \setminus Z)$, then $g(x) \in Y \setminus Z$ and $x \in g^{-1}(Y)$. Since $g(x) \notin Z$, we have $x \notin g^{-1}(Z)$. Therefore, $x \in g^{-1}(Y) \setminus g^{-1}(Z)$. For the converse, if $x \in g^{-1}(Y) \setminus g^{-1}(Z)$, then $g(x) \in Y$ and $g(x) \notin Z$, meaning $g(x) \in Y \setminus Z$. Therefore, $x \in g^{-1}(Y \setminus Z)$.

Therefore, $\text{Ker } f$ is closed in I . Let C be a connected component of $\text{Ker } f$. Since connected components are closed, we have that C is closed in $\text{Ker } f$. Then $\text{Ker } f \setminus C = K \cap \text{Ker } f$ for some open set $K \subseteq I$. Then $C = \text{Ker } f \setminus (K \cap \text{Ker } f)$ is the set of all $x \in I$ with $x \in \text{Ker } f$ and $x \notin K$. This is precisely the set $(I \setminus K) \cap \text{Ker } f$. Since K is open, $I \setminus K$ is closed in I . Since $\text{Ker } f$ is closed in I , we have that C is an intersection of closed sets, and thus is closed in I .

Therefore, C is a closed, connected subspace of the unit interval I . Closed connected subspaces of intervals in $\mathbb{R}$ are precisely closed intervals. Therefore, C is of the form $[a, b]$. Since every (sub)space is a union of its connected components, we have

$$\text{Ker } f = \bigcup_{j \in J} C_j = \bigcup_{j \in J} [a_j, b_j]$$

for $\{C_j\}_{j \in J} = \{[a_i, b_i]\}_{j \in J}$ the collection of distinct connected components of $\text{Ker } f$. Furthermore, the C_j s are all disjoint. This will be quite important.

We are now prepared to define the map $g : I \rightarrow \mathbb{R}$ such that $gf = 0$. We define g piecewise as follows. Define $g_0 : I \setminus \text{Ker } f \rightarrow \mathbb{R}$ as $g_0(x) = 0$. For each $j \in J$, define $g_j : [a_j, b_j] \rightarrow \mathbb{R}$ as the map

$$g_j(x) = \begin{cases} x - a_j, & a_j \leq x \leq \frac{b_j - a_j}{2} \\ b_j - x, & \frac{b_j - a_j}{2} \leq x \leq b_j \end{cases}.$$

We will define

$$g(x) = \begin{cases} g_j(x), & x \in C_j \\ g_0(x), & x \in I \setminus \text{Ker } f \end{cases}.$$

It is not clear that this map is continuous. We make use of the *pasting lemma*.

Lemma 8.1. *Let X, Y be closed subspaces of a space A such that $X \cup Y = A$. If $f : A \rightarrow Z$ is a function to a space Z such that the restrictions $f|_X, f|_Y$ are continuous $X, Y \rightarrow Z$, then f is continuous.*

Proof. Let U be a closed subset of Z . Since f is continuous on X and Y , we have that the preimages $f^{-1}(U) \cap X$ and $f^{-1}(U) \cap Y$ are closed in X and Y , since preimages of closed sets under continuous maps are closed. They are thus closed in A since X and Y are closed, and thus $f^{-1}(U) = (f^{-1}(U) \cap X) \cup (f^{-1}(U) \cap Y)$ is a (finite) union of closed sets, and thus closed. We have

$$(f^{-1}(U) \cap X) \cup (f^{-1}(U) \cap Y) = f^{-1}(U) \cap (X \cup Y) = f^{-1}(U) \cap A = f^{-1}(U).$$

Therefore, preimages of closed sets are closed under f .

Let $V \subseteq Z$ be open. Then $V = Z \setminus U$ for some closed U . The preimage $f^{-1}(U)$ is thus closed, so

$$f^{-1}(V) = f^{-1}(Z \setminus U) = f^{-1}(Z) \setminus f^{-1}(U)$$

is a complement of a closed set, and thus is open. Therefore, preimages of open sets are open, and f is continuous. ■

Define g_J as the map $g_J(x) = g_j(x)$ whenever $x \in C_j$. Since the connected components are disjoint, the map is well-defined. We would like to combine g_0 and g_J and apply lemma 8.1, but unfortunately $I \setminus \text{Ker } f$ is not necessarily closed. We thus extend g_0 to the closure $\overline{I \setminus \text{Ker } f}$. If g_J is continuous and agrees with g_0 on the intersection of the domains, then we will have a well-defined continuous map $g : I \rightarrow \mathbb{R}$ by lemma 8.1.

We start by showing that g_J is continuous. Since each subspace C_j is disjoint, there is a clear homeomorphism with the coproduct

$$\bigcup_{j \in J} C_j \cong \coprod_{j \in J} C_j.$$

I proved in a recent MA721 homework that (small) coproducts exist in $\mathcal{T}\text{op}$, and have the same point-set construction as in $\mathcal{S}\text{et}$. By universality of coproduct, it suffices to show that each $g_j : C_j \rightarrow \mathbb{R}$ is continuous. Here we once again apply lemma 8.1. Given $C_j = [a_j, b_j]$, write $c_j = a_j + \frac{b_j - a_j}{2}$ for the midpoint of $[a_j, b_j]$. Then $C_j = [a_j, c_j] \cup [c_j, b_j]$. Recall our definition

$$g_j(x) = \begin{cases} x - a_j, & x \in [a_j, c_j] \\ b_j - x, & x \in [c_j, b_j] \end{cases}.$$

Note that since subtraction or addition by a constant is continuous, and since the identity $x \mapsto x$ is continuous, restricting along $g_j|_{[a_j, c_j]}$ yields a composite of continuous functions, and thus is continuous. Similarly, since addition by a constant and the inverse of the identity are continuous, the restriction $g_j|_{[c_j, b_j]}$ is also a composite of continuous functions, and is thus continuous. Since $[a_j, c_j], [c_j, b_j]$ cover C_j and are closed sets, by lemma 8.1 it suffices to show these definitions agree on $c_j = a_j + \frac{b_j - a_j}{2}$; that is, we are showing that g_j is well-defined. Indeed,

$$\begin{aligned} a_j + \frac{b_j - a_j}{2} - a_j &= a_j + \frac{b_j - 3a_j}{2} = \frac{b_j - a_j}{2}, \\ b_j - a_j - \frac{b_j - a_j}{2} &= \frac{2b_j - 2a_j - b_j + a_j}{2} = \frac{b_j - a_j}{2}. \end{aligned}$$

Therefore, g_j is continuous, and by universality g_J is continuous. It remains only to be shown that g_J and g_0 agree on the intersection. By our knowledge of boundaries of open sets in $\mathbb{R}$, it suffices to show that g_J vanishes on the boundary of each connected component. Indeed,

$$g_j(a_j) = a_j - a_j = 0, \quad g_j(b_j) = b_j - b_j = 0.$$

This gives us our map $g : I \rightarrow \mathbb{R}$. We immediately see that $g \cdot f = 0$, since whenever $f(x) \neq 0$, we have $g(x) = 0$ by construction. The question is whether $g = 0$ or not. g is only nonvanishing on connected components of the kernel of f . If each connected component is a singleton, meaning f only vanishes at individual points, then the entirety of the connected component is on the boundary. Since g vanishes on the boundary, it vanishes along the entirety of the component, and thus $g = 0$. Therefore, we require that at least one connected component of $\text{Ker } f$ be larger than a singleton. A cleaner way to phrase this is that f vanishes on an open interval, since it necessarily vanishes on a closed interval by continuity and closure of $\{0\}$, and the open interval $(a, b) \subseteq [a, b]$ condition simply states that $[a, b]$ is not a singleton. Therefore, f is a zero divisor if and only if it vanishes on an open interval. ■

Proof of part 3. Units of $\mathcal{R}$ are precisely functions f which are everywhere nonvanishing. Suppose that f vanishes at some x , and suppose for contradiction that there is $g \in \mathcal{R}$ with $g \cdot f = 1$. Then $(g \cdot f)(x) = 1$. We have

$$g(x)f(x) = g(x)0 = 1,$$

a contradiction. Therefore, f is not a unit.

Now let f be everywhere nonvanishing. Division by a nonzero quantity is continuous. Write $(-)^{-1}$ for the map $\mathbb{R}_{\neq 0} \rightarrow \mathbb{R}$ given by $x \mapsto \frac{1}{x}$. Since f vanishes nowhere, it factors through $\mathbb{R}_{\neq 0}$, meaning that there is a map $\frac{1}{f} = (-)^{-1} \circ f : I \rightarrow \mathbb{R}$ given by $x \mapsto \frac{1}{f(x)}$. This is easily an inverse to f :

$$\left(\frac{1}{f} \cdot f\right)(x) = \frac{1}{f(x)}f(x) = 1.$$

Therefore, f is a unit. ■

Problem 9

Problem 9.1. Let $a \in R$ be nilpotent for R a commutative ring. Show that

1. a is either 0 or a zero-divisor,
2. ra is nilpotent for $r \in R$,
3. $1 + a$ is a unit in R , and
4. $u + a$ is a unit for all $u \in R^\times$.

Proof. (a) 0 is clearly nilpotent, so let $a \neq 0$ be nilpotent. Then $a^m = 0$ for some m , so we have that $aa^{m-1} = 0$. Take m to be the smallest positive number such that this is true, which is greater than 1 since $a = a^1 \neq 0$. Therefore, $a^{m-1} \neq 0$, and thus a is a zero divisor.

(b) Since R is commutative, $(ra)^m = r^m a^m = r^m 0 = 0$.

(c) First, note that

$$a + (-1 \cdot a) = (1 \cdot a) + (-1 \cdot a) = (1 - 1)a = 0a = 0.$$

Therefore, since R is an abelian group, we have $-1 \cdot a = -a$ by uniqueness of inverses. Now given $n \geq 1$, note that

$$(1 - r) \sum_{i=0}^{n-1} r^i = \left(\sum_{i=0}^{n-1} r^i \right) - \left(\sum_{i=0}^{n-1} r^{i+1} \right) = 1 - r^n.$$

The first equality follows by commutativity and distributivity of R , and the second by the fact that most terms in the summations cancel.

By (b), $-1 \cdot a = -a$ is nilpotent, meaning $(-a)^m = 0$ for some $m \geq 0$. Applying the expansion identity we found,

$$1 - (-a)^m = (1 - (-a)) \sum_{i=0}^{m-1} (-a)^i.$$

By our knowledge of groups, $-(-a) = a$. Additionally, since $(-a)^m = 0$, we have

$$1 = 1 - 0 = 1 - (-a)^m = (1 + a) \sum_{i=0}^{m-1} (-a)^i.$$

Therefore,

$$1 = (1 + a) \sum_{i=0}^{m-1} (-a)^i.$$

By commutativity of R , the summation on the right provides a two-sided inverse for $1 + a$.

(d) Note that for a unit $u \in R^\times$, there exists $u^{-1} \in R^\times$, and

$$u + a = u(1 + u^{-1}a), .$$

Since $R^\times$ is closed under multiplication and u is a unit, it suffices to show $1 + u^{-1}a$ is a unit. By (c), $1 + u^{-1}a$ is a unit if $u^{-1}a$ is nilpotent, and by (b) it is indeed nilpotent since a is nilpotent. ■

Problem 10

Problem 10.1. *Show that any finite integral domain is a field.*

Proof. It suffices to show all nonzero elements have inverses. Let R be an integral domain, and $r \in R$ nonzero. Write R^* for $R \setminus \{0\}$. We claim that the function $L_r : R^* \rightarrow R^*$ given by left multiplication $a \mapsto ra$ is a bijection. The fact that the codomain can be taken to be R^* follows since there are no zero divisors, and 0 is not in the domain.

Since R is finite, R^* is finite, and thus it suffices to show either injectivity or surjectivity. The former is easier, so suppose that there are $a, b \in R^*$ such that $ra = rb$. Then $ra - rb = 0$, and thus $r(a - b) = 0$. Since R is an integral domain, it has no zero divisors, and since $r \neq 0$, we must have $a - b = 0$. Therefore, $a = b$, and L_r is bijective.

From surjectivity of L_r , we see that there exists a (unique) $a \in R^*$ such that $ra = 1$. Therefore, r has a right-inverse. Since integral domains are commutative, we have $ar = 1$, and thus a is also a left-inverse of r . Therefore, r has an inverse element for each $r \in R^*$, meaning r is a unit. Since all nonzero elements of R are units, by definition it is a field. ■

Problem 11

Problem 11.1. *Show that if $n \geq 2$, then $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime.*

Proof. Let n be prime. By problem 10.1, it suffices to show $\mathbb{Z}/n\mathbb{Z}$ is an integral domain, since it is finite. Since $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring for all n , it suffices to show it has no zero divisors. Suppose for contradiction that there are nonzero $[a], [b] \in \mathbb{Z}/n\mathbb{Z}$ with $[a] \cdot [b] = [ab] = [0]$. Then by definition, we must have $n \mid ab$. However, since n is prime, it follows that $ab = n^k$ for some k . Since n is prime and $a, b \mid ab = n^k$, we must have $a = n^p$ and $b = n^q$ for some p, q . It follows that $a, b \geq n$, and in particular $n \mid a, b$. We thus have $a, b \in [n]$, and thus $[a] = [b] = [0]$, a contradiction. Therefore, $\mathbb{Z}/n\mathbb{Z}$ has no zero divisors, and is thus a field.

We prove the converse by contrapositive. Let n not be prime. Then there exist $n > a, b \geq 2$ such that $n = ab$. We then have

$$[a] \cdot [b] = [a \cdot b] = [n] = [0]$$

because $n \equiv 0 \pmod{n}$. Additionally, since $a, b < n$ and $a, b \neq n$, we must have $a, b \not\equiv n \pmod{n}$, meaning $[a], [b] \neq [0]$. Therefore, there exists at least one zero divisor, and $\mathbb{Z}/n\mathbb{Z}$ cannot be a field since a field has no zero divisors. ■